

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,  
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS2

SUBJECT: Biology

## **TOPIC: The Endocrine System**

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### **Introduction**

- The endocrine system consists of glands that secrete hormones directly into the blood. Hormones are chemical messengers that travel to target organs and regulate their activities — growth, metabolism, reproduction and homeostasis.

### **Major endocrine glands and their hormones**

- **Pituitary (master gland)** — growth hormone (GH), and hormones controlling other glands.
- **Thyroid (neck)** — thyroxine, controlling metabolism; deficiency causes goitre and cretinism.
- **Pancreas (islets of Langerhans)** — insulin and glucagon, regulating blood sugar; insulin deficiency causes diabetes mellitus.
- **Adrenal glands (on the kidneys)** — adrenaline, preparing the body for action (fight or flight); aldosterone and cortisol.
- **Gonads** — ovaries (oestrogen, progesterone) and testes (testosterone), controlling the secondary sexual characters.
- **Parathyroid** — parathormone, controlling calcium.

### **Hormones vs nerves**

- Hormones act slowly but for a long time, and travel in the blood; nerve impulses are fast but brief. Both work together to keep the body coordinated.

### **CLASS EVALUATION**

1. Define a hormone.
2. Name the gland that produces insulin and state its function.
3. What is the effect of thyroxine deficiency?
4. Distinguish between hormonal and nervous coordination.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS2

SUBJECT: Biology

## **TOPIC: Reproduction in Plants**

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### **Introduction**

- Reproduction is the production of new individuals of the same species. Flowering plants reproduce sexually through flowers; the parts of the flower are adapted for pollination and fertilisation.

### **Structure of a flower**

- **Calyx** — sepals, protect the bud.
- **Corolla** — petals, brightly coloured to attract insects.
- **Androecium** — the stamens (filament + anther) that produce pollen.
- **Gynoecium** — the pistil: stigma (receives pollen), style and ovary (with ovules).
- A flower with all four whorls is complete; with both stamens and pistil it is bisexual (e.g. hibiscus); a flower with only stamens is male (staminate), only pistil female (pistillate).

### **Pollination and fertilisation**

- Pollination is the transfer of pollen from the anther to the stigma — by insects (entomophily), wind (anemophily), birds, water or bats. After pollination a pollen tube grows down the style; the male nucleus fuses with the egg nucleus (fertilisation). The ovule becomes the seed and the ovary the fruit.

### **CLASS EVALUATION**

1. Label the parts of a flower.
2. Differentiate between insect- and wind-pollinated flowers.
3. Define fertilisation in plants.
4. What becomes of the ovule and ovary after fertilisation?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS2

SUBJECT: Biology

## TOPIC: Pollination, Seeds & Fruits

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### Introduction

- This week continues plant reproduction: the agents of pollination, seed and fruit formation, dispersal, and germination — the processes that spread and establish plants.

### Agents of pollination

- **Insects** — large, brightly coloured, scented flowers with nectar (hibiscus, sunflower).
- **Wind** — small, dull flowers with much light pollen, feathery stigmas (maize, grasses).
- **Water** — floating plants (water lily).
- Self-pollination (same flower or plant) and cross-pollination (different plants of the same species) — cross-pollination gives more variation.

### Fruit and seed dispersal

- Fruits protect the seeds and help dispersal: by wind (light, winged fruits — the silk cotton, dandelion), by animals (sweet, hooked or burred fruits — mango, Desmodium), by water (coconut), by explosive mechanism (the bean pod, balsam), and by man (cultivation).

### Germination

- Conditions: water, warmth and oxygen (some seeds need light or darkness). Types: **epigeal** — the cotyledons rise above the soil (bean); **hypogeal** — the cotyledons remain below the soil (maize, pea). The radicle (root) emerges first, then the plumule (shoot).

### CLASS EVALUATION

1. State four agents of seed dispersal and one example each.
2. Differentiate between self- and cross-pollination.
3. What conditions are needed for germination?
4. Distinguish between epigeal and hypogeal germination.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS2

SUBJECT: Biology

## **TOPIC: Reproduction in Humans**

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### **Introduction**

- Human reproduction is sexual: a sperm (male gamete) fuses with an egg (female gamete) to form a zygote. The reproductive systems are adapted for gamete production, fertilisation, pregnancy and birth.

### **The male system**

- Testes — produce sperm and testosterone; epididymis — stores sperm; vas deferens — carries sperm; seminal vesicles and prostate gland — add seminal fluid; the penis — transfers sperm. Sperm: head (nucleus), middle piece (mitochondria for energy) and tail (movement).

### **The female system**

- Ovaries — produce eggs and oestrogen/progesterone; oviduct (fallopian tube) — the site of fertilisation; uterus (womb) — the site of implantation and development; cervix; vagina — receives sperm and births the baby. The menstrual cycle lasts about 28 days: menstruation, thickening of the uterine lining, ovulation (about day 14) and the luteal phase.

### **Fertilisation to birth**

- Fertilisation in the oviduct → zygote → cleavage → implantation in the uterine wall → the placenta forms (nutrients, gases and wastes exchanged; the umbilical cord connects the fetus); after about nine months (gestation) the baby is born. Twins: identical (one zygote) or fraternal (two eggs).

### **CLASS EVALUATION**

1. Describe the functions of the testes and the ovaries.
2. Where does fertilisation occur?
3. What is the function of the placenta?
4. State the stages of the menstrual cycle.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS2

SUBJECT: Biology

## **TOPIC: Pregnancy, Birth & Growth**

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### **Introduction**

- This week follows human development: the changes of pregnancy, the stages of childbirth, and the growth and development of the child from birth to adulthood.

### **Pregnancy and its care**

- Signs: a missed period, nausea, enlarged breasts; the fetus develops the heart, brain and limbs in the first trimester. Antenatal care: regular clinic visits, a balanced diet (extra protein, iron, folic acid and calcium), rest, exercise, no alcohol or smoking, and immunisation (tetanus). The mother gains weight; the fetus moves in the fourth month.

### **Birth (parturition)**

- Stages: (1) labour — the uterus contracts, the cervix dilates, the waters break; (2) delivery — the baby emerges head first; (3) the placenta (afterbirth) is expelled. The cord is cut; the baby's first breath inflates the lungs.

### **Growth and development**

- Growth is an increase in size and mass; development is the acquisition of skills and functions. Infancy (0–1), childhood, adolescence (puberty — the physical changes controlled by sex hormones), adulthood. Growth curves of height show rapid infant growth, steady childhood growth and the adolescent spurt — then a plateau. Growth is measured by weighing and measuring, and compared with norms.

### **CLASS EVALUATION**

1. State three rules of antenatal care.
2. Name the three stages of birth.
3. Differentiate between growth and development.
4. What changes mark puberty in boys and girls?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS2

SUBJECT: Biology

## **TOPIC: Growth & Development (Measurement & Patterns)**

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### **Introduction**

- This week consolidates growth: how it is measured, the patterns shown by growth curves, and the factors that affect human growth and development.

### **Measuring growth**

- By mass (weight), height/length, and the head circumference of infants; against standard percentile charts; by weighing at the clinic and keeping the child's growth record. In plants, growth is measured by height, leaf number and dry mass.

### **Patterns**

- Human growth is not uniform: rapid in infancy, slow in childhood, a spurt at adolescence, then a plateau. Plant growth is sigmoid (S-shaped): the lag phase (early, slow), the log phase (rapid), the deceleration phase and the stationary phase.

### **Factors affecting growth**

- Heredity; hormones (GH, thyroxine, sex hormones); nutrition (food and vitamins); health (disease, parasites); environment (exercise, rest, sunlight); and the care given to the child. Malnutrition and infections during childhood permanently stunt growth — the reason for immunisation, good food and the growth-monitoring clinic.

### **CLASS EVALUATION**

1. State two ways of measuring growth.
2. Sketch the sigmoid growth curve of a plant and label its phases.
3. Mention three factors that affect human growth.
4. Why is the adolescent growth spurt important?